pollen-classifier

April 30, 2025

[3]:
```
# This Python 3 environment comes with many helpful analytics libraries␣
↪installed
# It is defined by the kaggle/python Docker image: https://github.com/kaggle/
↪docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will list␣
↪all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/) that␣
↪gets preserved as output when you create a version using "Save & Run All"
# You can also write temporary files to /kaggle/temp/, but they won't be saved␣
↪outside of the current session
```

```
/kaggle/input/pollen-grain-image-classification/combretum_11.jpg
/kaggle/input/pollen-grain-image-classification/faramea_15.jpg
/kaggle/input/pollen-grain-image-classification/myrcia_34.jpg
/kaggle/input/pollen-grain-image-classification/croton_22.jpg
/kaggle/input/pollen-grain-image-classification/arecaceae_16.jpg
/kaggle/input/pollen-grain-image-classification/combretum_08.jpg
/kaggle/input/pollen-grain-image-classification/senegalia (18).jpg
/kaggle/input/pollen-grain-image-classification/syagrus_32.jpg
/kaggle/input/pollen-grain-image-classification/myrcia_22.jpg
/kaggle/input/pollen-grain-image-classification/qualea_13.jpg
/kaggle/input/pollen-grain-image-classification/qualea_23.jpg
/kaggle/input/pollen-grain-image-classification/anadenanthera_18.jpg
/kaggle/input/pollen-grain-image-classification/mabea_17.jpg
/kaggle/input/pollen-grain-image-classification/syagrus_14.jpg
/kaggle/input/pollen-grain-image-classification/tridax_28.jpg
```

```
/kaggle/input/pollen-grain-image-classification/arrabidaea_17.jpg
/kaggle/input/pollen-grain-image-classification/hyptis_21.jpg
/kaggle/input/pollen-grain-image-classification/arrabidaea_21.jpg
/kaggle/input/pollen-grain-image-classification/arrabidaea_02.jpg
/kaggle/input/pollen-grain-image-classification/faramea_24.jpg
/kaggle/input/pollen-grain-image-classification/syagrus_25.jpg
/kaggle/input/pollen-grain-image-classification/faramea_17.jpg
```

[1]:
```
! [ -e /content ] && pip install -Uqq fastbook
import fastbook
fastbook.setup_book()
from fastbook import *
from fastai.vision.widgets import *
```

```
                         719.8/719.8 kB
22.7 MB/s eta 0:00:00
                         124.1/124.1 kB
7.8 MB/s eta 0:00:00
                         183.9/183.9 kB
12.1 MB/s eta 0:00:00
                         246.9/246.9 kB
16.3 MB/s eta 0:00:00
                         363.4/363.4 MB
4.4 MB/s eta 0:00:000:00:0100:01
                         664.8/664.8 MB
2.2 MB/s eta 0:00:00:00:01m00:01
                         211.5/211.5 MB
3.7 MB/s eta 0:00:000:00:0100:01
                         56.3/56.3 MB
3.2 MB/s eta 0:00:000:00:0100:01
                         127.9/127.9 MB
12.8 MB/s eta 0:00:00:00:0100:01
                         207.5/207.5 MB
8.0 MB/s eta 0:00:000:00:0100:01
                         21.1/21.1 MB
78.3 MB/s eta 0:00:00:00:0100:01
                         1.6/1.6 MB
53.6 MB/s eta 0:00:00
```

```
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of the
following dependency conflicts.
gcsfs 2024.10.0 requires fsspec==2024.10.0, but you have fsspec 2024.12.0 which
is incompatible.
bigframes 1.36.0 requires rich<14,>=12.4.4, but you have rich 14.0.0 which is
incompatible.
pylibcugraph-cu12 24.12.0 requires pylibraft-cu12==24.12.*, but you have
pylibraft-cu12 25.2.0 which is incompatible.
pylibcugraph-cu12 24.12.0 requires rmm-cu12==24.12.*, but you have rmm-cu12
25.2.0 which is incompatible.
```

[2]:
```
from fastai.vision.all import *
```

[5]:
```
!ls /kaggle/input
```

```
pollen-grain-image-classification
```

[6]:
```
from pathlib import Path
import shutil
from sklearn.model_selection import train_test_split
```

[7]:
```
input_dir = Path("/kaggle/input/pollen-grain-image-classification")
#  jpg
all_images = list(input_dir.glob("*.jpg"))
```

[8]:
```
base = Path("/kaggle/working/pollen-data")
train_dir = base/'train'
valid_dir = base/'valid'
```

[10]:
```
#
label_to_images = {}#
for img in all_images:
    label = img.stem.split('_')[0]#img.stem     ,split   ['anadenanthera', '1']
    label_to_images.setdefault(label, []).append(img)
    #setdefault key       key,append
```

[12]:
```
from sklearn.model_selection import train_test_split
from pathlib import Path
import shutil

base = Path("/kaggle/working/pollen-data")
```

```
train_dir = base/'train'
valid_dir = base/'valid'

for label, images in label_to_images.items():
    if len(images) < 2:
        print(f"   {label}    {len(images)}  ")
        continue  #

    train_imgs, valid_imgs = train_test_split(images, test_size=0.2,␣
 ↪random_state=42)

    for split, split_imgs in [('train', train_imgs), ('valid', valid_imgs)]:
        dest = base/split/label
        dest.mkdir(parents=True, exist_ok=True)
        for img in split_imgs:
            shutil.copy(img, dest/img.name)

print("     ")
```

```
  senegalia (18)    1
  senegalia (30)    1
  senegalia (25)    1
  senegalia (33)    1
  senegalia (16)    1
  urochloa (13)    1
  urochloa (30)    1
  urochloa (12)    1
  urochloa (29)    1
  urochloa (9)    1
  senegalia (2)    1
  urochloa (15)    1
  urochloa (33)    1
  senegalia (1)    1
  urochloa (10)    1
  urochloa (27)    1
  urochloa (34)    1
  urochloa (11)    1
  senegalia (27)    1
  senegalia (20)    1
  urochloa (2)    1
  senegalia (23)    1
  senegalia (28)    1
  urochloa (31)    1
  urochloa (20)    1
  senegalia (17)    1
  senegalia (35)    1
  urochloa (5)    1
  senegalia (34)    1
```

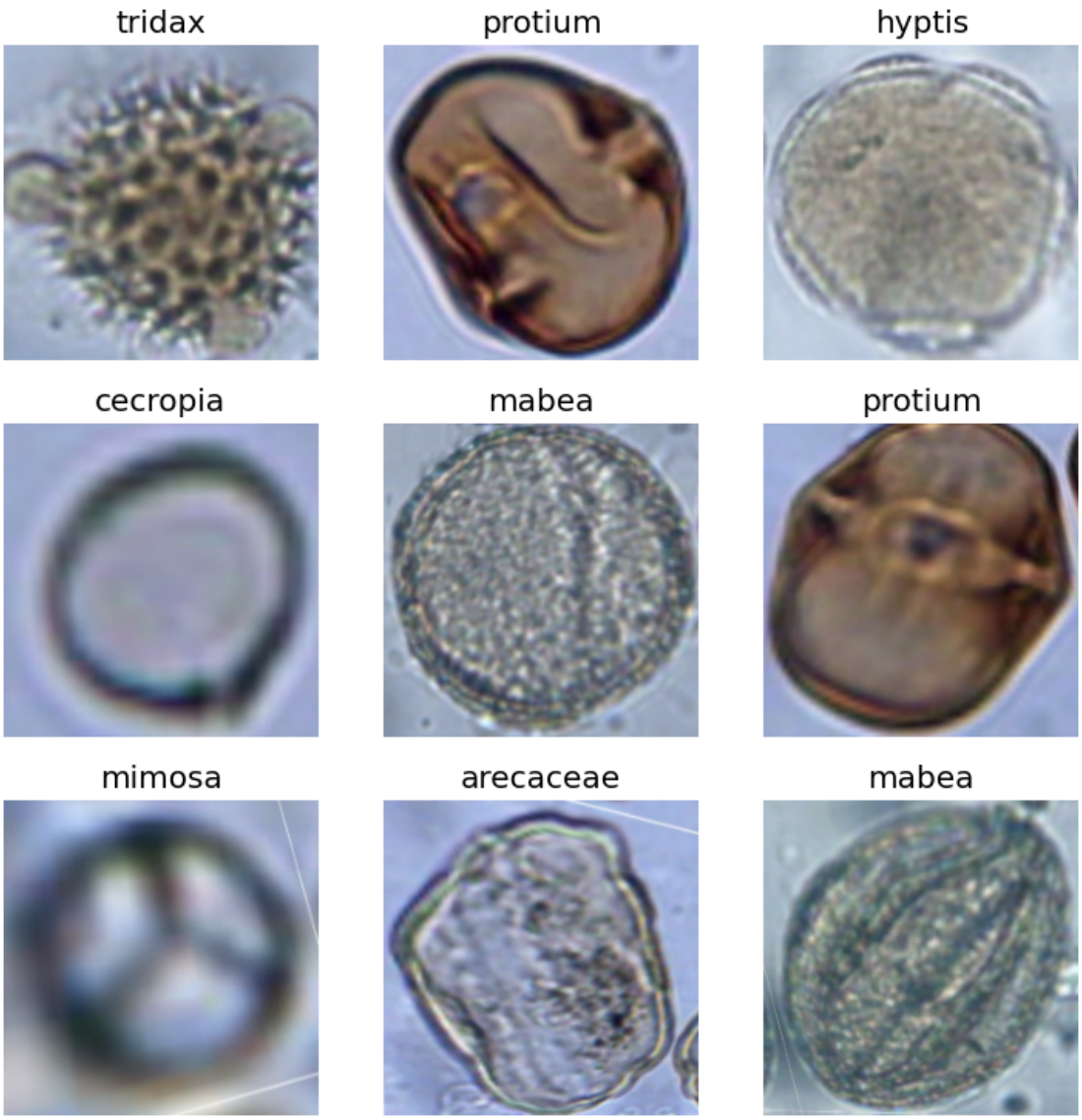
```
urochloa (18)    1
urochloa (23)    1
urochloa (4)    1
senegalia (22)    1
urochloa (7)    1
senegalia (32)    1
urochloa (1)    1
urochloa (21)    1
urochloa (17)    1
senegalia (29)    1
urochloa (28)    1
senegalia (24)    1
senegalia (5)    1
senegalia (7)    1
urochloa (16)    1
senegalia (12)    1
senegalia (21)    1
senegalia (15)    1
senegalia (3)    1
urochloa (25)    1
senegalia (8)    1
senegalia (19)    1
senegalia (26)    1
urochloa (14)    1
urochloa (26)    1
senegalia (6)    1
urochloa (6)    1
senegalia (11)    1
urochloa (19)    1
urochloa (32)    1
urochloa (3)    1
senegalia (10)    1
senegalia (4)    1
urochloa (8)    1
senegalia (9)    1
senegalia (13)    1
urochloa (35)    1
urochloa (22)    1
senegalia (31)    1
urochloa (24)    1
senegalia (14)    1
```

[14]:
```
from fastai.vision.all import *

dls = ImageDataLoaders.from_folder(
    base,
```

```
    train='train',
    valid='valid',
    item_tfms=Resize(224, method='squish'),  #       224x224
    batch_tfms=aug_transforms(),
    bs=32
)

dls.show_batch(max_n=9)
```



[15]:
```
learn = vision_learner(dls, resnet18, metrics=error_rate)
learn.fine_tune(4)
```

```
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to
/root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth
100%|      | 44.7M/44.7M [00:00<00:00, 147MB/s]
```

```
<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>
```

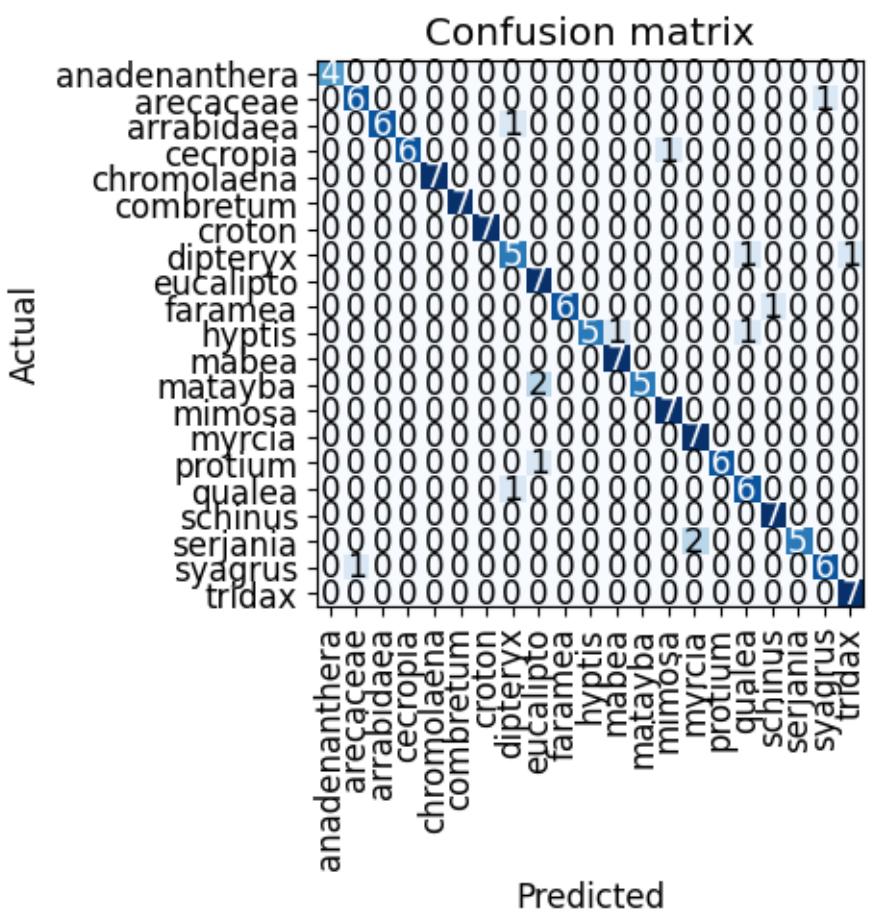
[16]:
```
interp = ClassificationInterpretation.from_learner(learn)
interp.plot_confusion_matrix()
```

```
<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>
```



[17]:
```
learn.export('/kaggle/working/pollen_model.pkl')
```